

Samukelo Mngadi

Business Analyst

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I turn business problems into requirements, and requirements into working products. I gather requirements, map processes and model data, and I build AI-assisted prototypes so stakeholders can try an idea instead of just reading about it.

AS A	business analyst,
I WANT	to prototype a solution alongside the requirements,
SO THAT	stakeholders can react to something real before development starts.

2+ yrs
of business analysis at UltimetaX and MICT SETA

4 in 4
AI tools built and shipped in four weeks

41
acceptance criteria written before coding Tower

81
automated tests keeping Tower honest

EXPERIENCE

Aug 2026 – Present

Data Engineering Intern

CAPACITI

- Data Engineering Programme: data pipelines, data modelling and analytics.
- CAPACITI x Clickatell AI Bootcamp (Sep–Oct 2026): built and shipped four AI projects in four weeks.
- Tower: wrote 10 user stories with 41 acceptance criteria, measured the classifier honestly (62.7% vs a 31.7% baseline) and designed the product around that result.

Mar 2025 – Feb 2026

Business Analyst Intern

MICT SETA

- Elicited, documented and validated business requirements; the link between stakeholders and developers.
- Wrote user stories with positive and negative acceptance criteria; turned requirements into development tasks.
- Mapped processes and system flows; built ERDs and data models.
- Designed wireframes in Figma and Visily; redesigned features after stakeholder feedback.
- Tested developer builds (staging, UAT, demos), reported defects, retested fixes, helped plan UAT.
- Scrum and Kanban: sprint planning, prioritisation, roadmaps and project plans in Microsoft Planner.

Jan 2023 – Jan 2024

Business Analyst

UltimetaX

- Gathered, recorded and verified business needs with stakeholders, aligned to operational objectives.
- Built end-to-end workflows, process diagrams and ERDs for system architecture, database planning and UI development.
- Wrote user stories with explicit acceptance criteria; helped with UI design and prototyping.
- Supported UAT planning, test cases and defect tracking; analysed gaps, risks and improvements across environments.

EDUCATION

Awarded Dec 2022

Diploma in ICT in Business Analysis

Durban University of Technology (DUT)

Passed all 31 modules with 6 distinctions. Top marks: Business Analysis Project I (86%), IT Project Management (84%), Business Analysis Project II (80%), Theory of Internal Auditing II (79%), Business Fundamentals II (78%), Financial Accounting I (75%).

Studied: Business Analysis II–III and four practical BA projects · Applications Development, Business Information Systems, Information Management, Computer Security · Business Fundamentals, Financial Accounting, Internal Auditing, Law · IT Project Management and ICT Professional Practice.

CASE STUDIES

AI BOOTCAMP · WEEK 4 · BUSINESS ANALYSIS, BUILD, TESTING

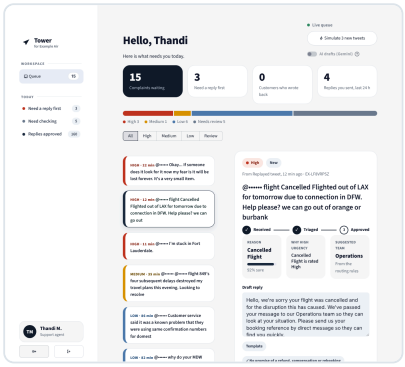
Tower: airline complaint triage assistant

Reads each customer complaint, masks personal details, suggests a reason and urgency, drafts a reply with Gemini, and leaves every decision to a person. Agents work an urgency-sorted queue and escalate to a team lead; customers follow progress on a status page. Several airlines share one app with four roles and fully separate data. Guardrails reject any draft that promises refunds, compensation or timeframes.

62.7% vs 31.7% baseline | 79% when ≥50% sure | 41 acceptance criteria | 81 tests

Python · Streamlit · scikit-learn · Gemini API · SQLite · Responsible AI · Role-based access

Live app · Code · Full case study



AI BOOTCAMP · WEEK 3 · DATA ANALYSIS, BUILD, REPORTING

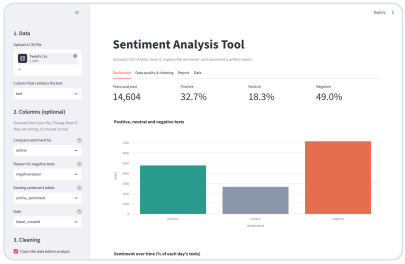
Sentiment analysis tool

Upload any CSV of comments, tweets or reviews and get a cleaned dataset with a change log, a sentiment dashboard and a written report. I tuned it one change at a time and kept only what improved results, creating a "customer messages" mode for complaint language.

14,604 tweets | 55% → 65.9% agreement | 51% → 69% negatives caught

Python · pandas · VADER · Streamlit · Plotly

Live app · Code · Full case study



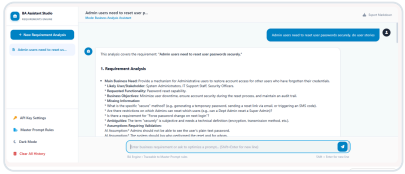
AI BOOTCAMP · WEEK 2 · PROMPT ENGINEERING, EVALUATION

AI business analyst assistant

Turns a one-line stakeholder requirement into a requirement analysis, user story, Given/When/Then acceptance criteria, test cases, clarification questions and a quality check. Built around a master prompt that forbids invented rules, then evaluated against a 7-point checklist on two real requirements.

Prompt engineering · Gemini API · React · TypeScript · Vite

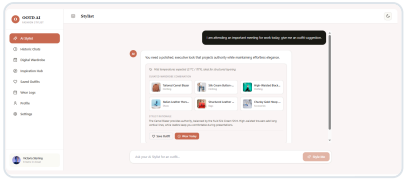
Code · Full case study



AI BOOTCAMP · WEEK 1 · PRODUCT DEFINITION, PROTOTYPING

#OOTD: AI personal stylist

An AI stylist that builds outfits only from the clothes you own, for any occasion, and explains why each look works. I wrote the product definition first (screens, closet categories, styling rules), then prototyped it in Google AI Studio.



Product definition · Google AI Studio · Gemini · React · TypeScript

Code · Full case study

MICT SETA

Case management system

Took a vague "we need a dashboard" brief, consulted other departments, and redesigned it around their feedback. Wrote user stories with acceptance criteria, tested features, logged bugs and drew up a sprint roadmap.

MICT SETA

Onboarding & verification platform

Requirements for employee, supplier and beneficiary onboarding with identity (KYC/AML), address, criminal, financial, citizenship and SAQA checks, plus HR dashboards, leave and PMDS. Researched verification API integrations.

SKILLS

Business analysis

- Requirements elicitation & validation
- User stories, Given/When/Then criteria
- Process maps, system flows, journeys
- ERDs and data models
- Stakeholder consultation

Delivery & testing

- Test cases, staging, UAT, demos
- Defect reporting and retesting
- Scrum, Kanban, SDLC
- Sprint planning, roadmaps
- Wireframes and UI concepts

Building with AI & data

- Prompt engineering & evaluation (Gemini)
- Python, pandas, Streamlit, SQLite
- scikit-learn text classification
- React & TypeScript basics, Git
- Responsible AI: human approval, guardrails

TOOLS I USE

PLANNING & DELIVERY
Jira · Microsoft Planner · Word · Excel · PowerPoint

DESIGN & PROTOTYPING
Figma · Visily · Google AI Studio

DATA & ANALYSIS
Python · Jupyter Notebook · MySQL · SQLite · pandas · scikit-learn · Highcharts · Plotly · Kaggle

AI & APP BUILDING
Gemini API · Streamlit · React · TypeScript · Vite · Node.js

VERSION CONTROL, CLOUD & HOSTING
Git · GitHub · GitHub Actions · AWS S3 · Apache · Vercel · Streamlit Community Cloud

REFLECTION

"The AI was the easy part."

The hard work on Tower was everything around it: who may see what, what happens when the model is unsure, how a customer follows up, and how one airline's data stays separate from another's. Measuring the classifier honestly is what led me to design it to suggest and flag rather than decide. I take the same approach into requirements work: write down the edge cases and limits before building, and show stakeholders something real early.